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Section 5.2: Searching for Data

This section will take a look at the methods used to search for data that has been stored in a database.

You will learn how to use something called '**operators**' which are used to structure your searches.

Key Concepts of this section:

- # How to use '**logical operators**' to structure searches
- # How to use '**mathematical operators**' to structure searches

What are operators?

- # Operators are used when searching for data in a database.
- # Operators are used to give the computer '**specific instructions**' on what to search for.

Examples of commonly used operators:



Click for Example of a
Logical Operator



Click for Example of a
Mathematical Operator



Key Words:

Logical Operators ,
Mathematical Operators

Logical Operators

- ✚ AND
- ✚ OR
- ✚ BETWEEN AND
- ✚ LIKE * *

Mathematical Operators (used with number values)

- ✚ Equal To (=)
- ✚ Less Than (<)
- ✚ More Than (>)
- ✚ Less Than OR Equal To (<=)
- ✚ More Than OR Equal To (>=)

Examples of how Operators are used within Searches

Look at the table of **Names**, **Genders** and **Heights** below.

We will use the table to show 3 examples of how **logical** and **mathematical operators** could be used to search it.

Name	Gender	Height (m)
Mike	Male	1.8
Salma	Female	1.72
Robert	Male	1.65
Karen	Female	1.6
Frank	Male	1.7

Example 1: **AND** with = and >

If we wanted to search the table for the following data:

(Gender = **Male**) **AND** (Height is **>1.7**)

The search would bring up the following result:

NOTE!

= will find an exact match.

> means 'greater than'

Name	Gender	Height (m)
Mike	Male	1.8
Salma	Female	1.72
Robert	Male	1.65
Karen	Female	1.6
Frank	Male	1.7

Explanation!

The search is looking for data that match **BOTH** parts of the criteria. For example the people in the orange cells would be returned as a result because:

Mike = Male ✓

Height >1.7 ✓

Some of the other people matched one part of the criteria but not both.
Some of the people did not match any of the criteria at all.

Example 2 : **OR** with = and <

If we wanted to search the table to find a different set of data such as:

(Gender = **Female**) **OR** (Height is **<1.7**)

NOTE!

= will find an exact match.

< means 'less than'

The search would bring up the following result:

Name	Gender	Height (m)
Mike	Male	1.8
Salma	Female	1.72
Robert	Male	1.65
Karen	Female	1.6
Frank	Male	1.7

Explanation!

This search is looking for data that match **EITHER** side of the criteria. The people in the orange cells would be returned as a result because:

Salma	= Female	✓	Height <1.7	✗
Robert	= Female	✗	Height <1.7	✓
Karen	= Female	✓	Height <1.7	✓

If **EITHER** side of the search is matched, the data will be returned in the result.

Example 3 : BETWEEN, AND

If we wanted to search the table to find a different set of data such as:

Height **BETWEEN** 1.6 **AND** 1.7

The search would bring up the following result:

NOTE!
Between, AND searches only
work with 2 sets of numbers

Name	Gender	Height (m)
Mike	Male	1.8
Salma	Female	1.72
Robert	Male	1.65
Karen	Female	1.6
Frank	Male	1.7

Explanation!

This search is looking for data that lies **within a range**. The people in the orange cells would be returned as a result because:

Robert Height = 1.65



Karen Height = 1.6



Frank Height = 1.7



NOTE!

All of these heights are between 1.6 and 1.7 metres

Activity!

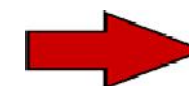
[Operator Searches - Task](#)

Click the above task and follow the instructions given on the sheet. Make sure that you complete all tasks on the sheet.



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